HOPE Final Project Showcase: Solar Charging Station

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Objective

Inspiration

The inspiration for this project came from seeing how much modern devices depend on electricity. Many people lose access to charging when they are outdoors, traveling, or in emergency situations. We wanted to build a portable power source that can work on its own and use solar energy to charge small electronics.

Our goal was to design a system that is compact, efficient, and easy to use. The system shows real-time information such as battery level and charging status.

Categories

Power

The system gathers energy from a 5V solar panel, which is protected by a resettable fuse and a Schottky diode to prevent backflow. The BQ24075 charger IC manages power flow, safely charging a 3.7V Li-Po battery. Additional components like the AP7333 LDO and TPS61023 boost converter regulate the voltages needed for different parts of the circuit.

Actuation

Once a device is actively charging, an OLED display will show both that the device plugged in is charging and what the charge of the LiPo battery is. To estimate the charge of the LiPo battery (since we are stepping down and stepping up the voltage in different parts of the circuit), a voltage divider approximation is used.

Computation

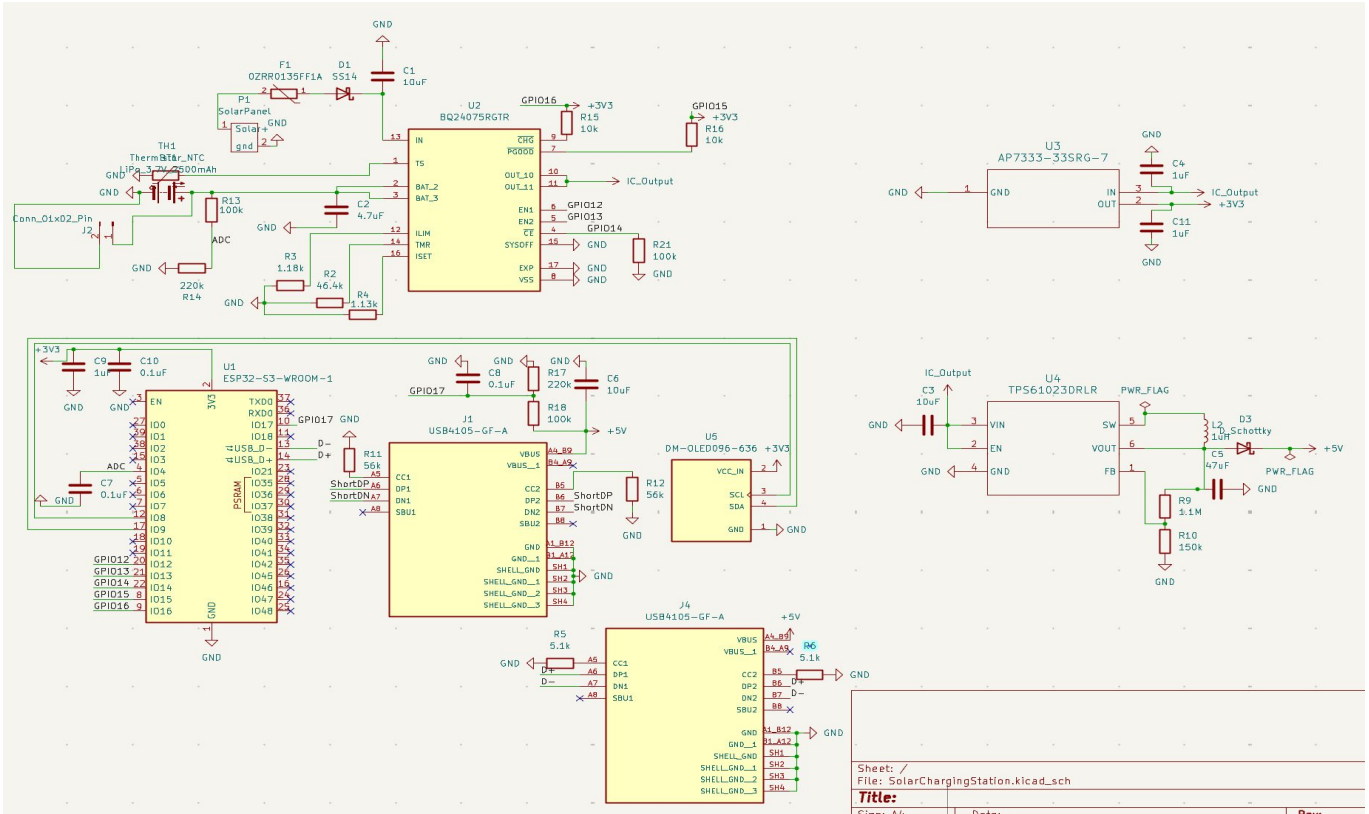
The ESP32 continuously measures battery voltage, monitors solar input, detects whether a device is plugged in and drawing current, and evaluates overall system status. Using this data, it communicates real-time information to users via a 0.96-inch OLED display. Examples include disabling charging when the battery is full or interpreting 5V output sag as a sign that an external device is charging.

Bill of Materials (BOM)

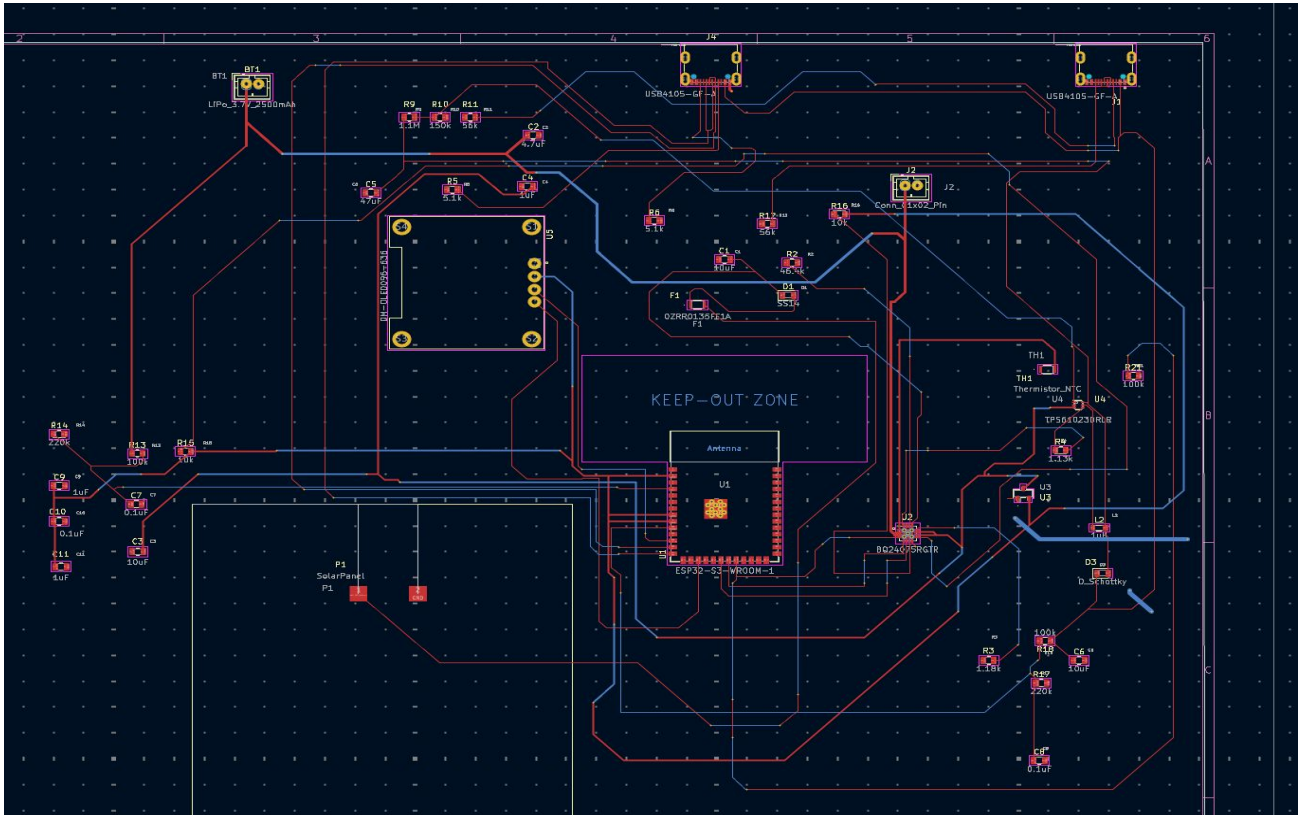
Here are the materials that we used to build the Solar Charging Station.

- Core Electronic
  - ESP32-S3-WROOM-1 Module Main
  - BQ24075 Battery Charger IC 1-cell Li-ion
  - TPS61023 Boost Converter (5V Output)
  - AP7333-3.3 LDO Regulator Provides stable
- Power Components
  - 3.7V 2500mAh Li-Po Battery
  - Adafruit 5V 1.2W Solar Panel
  - Resettable PTC Fuse (0ZRR0135FF1A)
  - Schottky Diode (SS14)
  - Inductor 1µH for boost converter
- Connectors
  - USB-C Power-Only Receptacle (x2)
  - JST-PH 2-pin Battery Connector
- Display
  - 0.96" OLED (SSD1306, I²C)
- Passive Components
  - Capacitors: 10µF, 4.7µF, 1µF, 0.1µF (various ceramic types)
  - Resistors: Feedback resistors (1.1MΩ, 150kΩ) Divider resistors (100kΩ, 220kΩ, 56kΩ)

Schematic


GND
F1
0ZRR0135FF1A
D1
SS14
C1
10uF
P1
SolarPanel
Solar+
gnd
U2
BQ24075RGTR
GPIO16
+3V3
R15
10k
GPIO15
R16
10k
TH1
Thermistor_NTC
LiPo_3.7V_2500mAh
R13
100k
ADC
C2
4.7uF
IC_Output
GPIO12
GPIO13
GPIO14
R21
100k
Conn_01x02_Pin
J2
R3
1.18k
R2
46.4k
R4
1.13k
220k
R14
U3
AP7333-33SRG-7
C4
1uF
C11
1uF
+3V3
U1
ESP32-S3-WROOM-1
C9
1uF
C10
0.1uF
C7
0.1uF
GPIO17
C8
0.1uF
R17
220k
C6
10uF
R18
100k
J1
USB4105-GF-A
R11
56k
ShortDP
ShortDN
R12
56k
U5
DM-OLED096-636
C3
10uF
U4
TPS61023DRLR
PWR_FLAG
L2
10uH
D3
D_Schottky
C5
47uF
R9
1.1M
R10
150k
J4
USB4105-GF-A
R5
5.1k
R6
5.1k
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Layout


KEEP-OUT ZONE
Antenna
ESP32-S3-WROOM-1
BQ24075RGTR
TPS61023DRLR
OZRR0135FE1A
USB4105-GF-A
Conn_1x02_Pin
Thermistor_NTC
LiPo_3.7V_2500mAh
SolarPanel

Assembled PCB

After we finished soldering all the parts, we were excited to try our PCB. Unfortunately, despite all code being written, hand-soldering the USB-C connector required to flash the microcontroller proved very challenging, and the microcontroller was not detectable.

However, even though the board didn't fully function, building it taught us a lot about designing circuits, making layouts, and solving real hardware problems, in addition to getting the hang of the general workflow for designing and creating PCBs.



Challenges and Setbacks

- **Schematic**
  - Understanding how the BQ24075 charger IC handled solar input, battery charging, and power-path control took time.
  - Designing correct resistor dividers for battery sensing and 5V detection required calculation and verification.
  - Integrating multiple regulators (LDO + boost converter) introduced complexity in ensuring stable voltage rails.
- **Layout**
  - Routing high-current paths for the charger and boost converter without noise issues or voltage drop.
  - Managing ground planes and vias to avoid interference between analog measurements and switching converters.
- **Assembly**
  - Hand-soldering the BQ24075 and TPS61023 was difficult due to small pads and thermal requirements.
  - Verifying that every connection, especially test points and USB-C pins, was properly soldered without hidden cold joints.
  - Troubleshooting non-responsive boards caused by subtle soldering errors (bridges, shorts, or incomplete connections).

Future Improvements

- More efficient solar panel
- A larger battery to extend usage time
- USB-C with fast-charging support
- Weather-resistance
- Wireless charging
- Better OLED display

Conclusion

In conclusion, our solar charging station shows how renewable energy, smart power control, and embedded computing can work together in a practical way. By combining a solar panel, a Li-Po battery, a charger IC, voltage regulators, and an ESP32 microcontroller, we created a portable system that delivers reliable 5V power. The system also monitors battery health, solar input, and device charging status in real time. This project highlights how careful circuit design and programming can turn simple components into a useful, self-sustaining solution that supports clean energy and everyday use.

Thank you for an amazing semester!!!